

Developing Python based Android Apps Using Kivy

Kivy is a cross-platform Python library, which can be used for the rapid development of apps that make use of innovative interfaces. This tutorial describes the installation of Kivy on various platforms and gives the code for building a few simple apps.

Nowadays, every business organisation strives to leverage its services with mobile apps. This permits wide coverage as well as global accessibility to the business services being offered. A number of software frameworks and libraries are distributed under different licences including proprietary, free and open source software. These software frameworks provide easy-to-use APIs (application programming interface) to the programmers, so that rapid and effective apps can be developed. With the use of mobile apps escalating each day, there is huge scope in the technology market to develop new apps for multiple uses and devices.

This area is being explored by the research and development teams of leading software development companies because of the dynamic way in which new products and services are rapidly being developed in the global market.

Kivy — a Python based cross-platform framework

Kivy is a powerful library based on Python for the development of mobile apps including the natural user interface (NUI). It is one of the effective cross-platform libraries that can run on iOS, Android, Raspberry Pi, Linux, Windows, MacOS X with the distribution protocol under free and open source software. It has been developed and is distributed by the Kivy organisation.

The features and elements of Kivy include:

- Support for multiple inputs including

the mouse, TUIO (Tangible User Interface Objects), the keyboard and multi-touch

- Powerful APIs for most mobile devices
- One application for multiple operating systems
- Support for networking protocols and remote login
- Support for a lot of widgets with multi-touch KV to design custom widgets

Installation and configuration of Kivy

Kivy can be installed on any operating system without any compatibility issues.

It can be downloaded from <http://kivy.org/#download>, and once this is done, the installation instructions are specified on the download page itself. For R&D purposes, the development version of Kivy for the latest releases and additions in the framework can be downloaded from GitHub using the following command:

```
$ git clone http://github.com/kivy/kivy
```

Classically, the development version of any software is used by programmers and testing engineers so that the latest release and updates can be analysed in real-time.

Installation of Kivy on Linux (Ubuntu/Debian based)

First, add the PPAs (Personal Package Archives) as follows:

stable builds:	<pre>\$ sudo add-apt-repository ppa:kivy-team/kivy</pre>
nightly builds:	<pre>\$ sudo add-apt-repository ppa:kivy-team/kivy-daily</pre>

Next, update the package list using a package manager. Now, install Kivy as follows.

Python2 Compatible version:

```
$ sudo apt-get install python-kivy
```

